Question	Answer	Marks	Partial Marks	Guidance
5(a)(i)(a)	1480	**1**		
5(a)(i)(b)	440	**2**	**M1** for [UQ =] 1600 soi or [LQ =] 1160 soi	M1 implied by e.g. 1600 – *k* Award M1 for answer 1600 or 1160
5(a)(ii)	$\frac{8}{80}$ oe	**2**	**M1** for 72 or 8 written	Only accept fraction, percentage and decimal equivalents (3sf or better) Ignore incorrect cancelling or conversion following a correct answer. 72 could be on diagram Allow M1 for $\frac{72}{80}$ oe but not for e.g. $\frac{1}{72}$ alone
5(b)	236 or 235.5 to 235.6	**3**	**M2** for $\frac{5.104}{1.3} \times 60$ oe or **M1** for $\frac{5.104}{\textit{their}\text{ time}}$	M2 for e.g. $5.104 \div \frac{78}{3600}$ oe , accept 0.0217 or 0.02166 to 0.02167 for $\frac{78}{3600}$ Condone 5104 used for 5.104 for M1 M1 implied by 3.93 or 3.926… For M1 *their* time could be e.g. 1.18, 78, 1m 18s or these values divided by 60 or (60 × 60) Ignore incorrect attempt at conversion for M1 If using hours allow 0.021 to 0.022 for *their* time
5(c)(i)	(160 – 120) × 0.2 [= 8]	**1**	with no errors seen	Accept 40 × 0.2 nfww Must show multiplication for the mark See AG

Question	Answer	Marks	Partial Marks	Guidance
5(c)(ii)	22, 36, 64, 30 seen	**B2**	**B1** for 2 or 3 correct frequencies or **M1** for three of 1.1 × (180 – 160), 1.8 × (200 – 180), 1.6 × (240 – 200) and 0.5 × (300 – 240) oe	Allow B2 or B1 if frequency products are used correctly in calculations with *their* midpoints to find Σfm.
	(8 × 140 + *their*22 × 170 + *their*36 × 190 + *their*64 × 220 + *their*30 × 270) ÷ 160	**M3**	**M1** for midpoints soi **M1** for Σfx, x in interval or boundary of interval **M1 dep on second M1** for $\Sigma fx \div 160$	*their* frequencies not fd's and must be integers Midpoints: 140, 170, 190, 220, 270 Condone one error/omission For M1 accept use of expressions such as 120 + (160 – 120)/2 etc for the midpoints. Allow one further error or omission fm: 1120, 3740, 6840, 14080, 8100. Total $\Sigma fm = 33880$
	211.75	**B1**		Condone $200 < v \leq 240$ after 211.75 seen Accept answers 212, 211.7 or 211.8 nfww See AG for alt methods

Additional Guidance Q5(c)(ii)

Question	Answer	Marks	Partial Marks	Guidance
ALTERNATIVE method using equal intervals of e.g. width 20				
5(c)(ii)	4, 4, 22, 36, 32, 32, 10, 10, 10	**B2**	**B1** for 7 or 8 correct frequencies or **M1** for correct products 20 × 1.1, 20 × 1.8, 20 × 1.6, 20 × 1.6, 20 × 0.5, 20 × 0.5, 20 × 0.5	Allow B2 or B1 if frequency products are used correctly in calculations with *their* midpoints to find Σfm.
	(4 × 130) + (4 × 150) + *their*22 × 170 + *their*36 × 190 + *their*32 × 210 + *their*32 × 230 + *their*10 × 250) + *their*10 × 270) + *their*10 × 290) ÷ 160	**M3**	**M1** for midpoints soi **M1** for Σfx, x in interval or boundary of interval **M1 dep on second M1** for Σfx ÷ 160	*their* frequencies not FD's and must be integers Midpoints: 130, 150, 170, 190, 210, 230, 250, 270, 290 Condone one error/omission For M1 accept use of expressions such as 120 + (140 – 120)/2 etc for the midpoints. Allow one further error or omission 520+600+3740+6840+6720+7360+2500+2700+2900 = 33880
	211.75	**B1**		Condone 200 < v ≤ 240 after 211.75 seen Accept answers 212, 211.7 or 211.8 nfww